COMPUTER VISION

LABSHEET 2

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SEC-C

#1. Import Required Libraries

```
import cv2
import numpy as np
import matplotlib.pyplot as plt
```

#2. Load a Grayscale Image

```
img = cv2.imread("image.jpg")

if img is None:
    print("Image not found!")
    exit()

print("Image shape:", img.shape)

img_color = cv2.imread("image.jpg")
img = cv2.cvtColor(img_color, cv2.COLOR_BGR2GRAY)
```

#3. Display Original Image and Histogram

```
plt.figure(figsize=(10, 4))

plt.subplot(1, 2, 1)
plt.imshow(img, cmap="gray")
plt.title("Original Image")
plt.axis("off")

plt.subplot(1, 2, 2)
plt.hist(img.ravel(), bins=256, range=[0, 256])
plt.title("Original Histogram")
plt.xlabel("Pixel Intensity")
plt.ylabel("Frequency")

plt.tight_layout()
plt.show()

#4. Perform Contrast Stretching4. Perform Contrast Stretching
min_val = np.min(img)
max_val = np.max(img)

contrast_stretched = ((img - min_val) * 255 /
                      (max_val - min_val)).astype(np.uint8)

plt.imshow(contrast_stretched, cmap="gray")
plt.title("Contrast Stretched Image")
```

```
plt.axis("off")
plt.show()

#5. Histogram Equalization
hist_equalized = cv2.equalizeHist(img)

plt.figure(figsize=(10, 4))

plt.subplot(1, 2, 1)
plt.imshow(hist_equalized, cmap="gray")
plt.title("Histogram Equalized")
plt.axis("off")

plt.subplot(1, 2, 2)
plt.hist(hist_equalized.ravel(), bins=256, range=[0, 256])
plt.title("Equalized Histogram")
plt.xlabel("Pixel Intensity")
plt.ylabel("Frequency")

plt.tight_layout()
plt.show()

#6. Apply CLAHE
clahe = cv2.createCLAHE(
    clipLimit=2.0,
    tileGridSize=(8, 8)
```

```
)

clahe_img = clahe.apply(img)
plt.figure(figsize=(10, 4))

plt.subplot(1, 2, 1)
plt.imshow(img, cmap="gray")
plt.title("Original")
plt.axis("off")

plt.subplot(1, 2, 2)
plt.imshow(clahe_img, cmap="gray")
plt.title("CLAHE Enhanced")
plt.axis("off")

plt.tight_layout()
plt.show()

#7. Compare All Histograms
plt.figure(figsize=(12, 8))

plt.subplot(3, 1, 1)
plt.hist(img.ravel(), bins=256, range=[0, 256])
plt.title("Original Histogram")

plt.subplot(3, 1, 2)
```

```
plt.hist(hist_equalized.ravel(), bins=256, range=[0, 256])
plt.title("Histogram Equalized")

plt.subplot(3, 1, 3)
plt.hist(clahe_img.ravel(), bins=256, range=[0, 256])
plt.title("CLAHE Histogram")

plt.tight_layout()
plt.show()

#8. Display All Results

plt.figure(figsize=(12, 8))

plt.subplot(2, 2, 1)
plt.imshow(img, cmap="gray")
plt.title("Original")
plt.axis("off")

plt.subplot(2, 2, 2)
plt.imshow(contrast_stretched, cmap="gray")
plt.title("Contrast Stretching")
plt.axis("off")

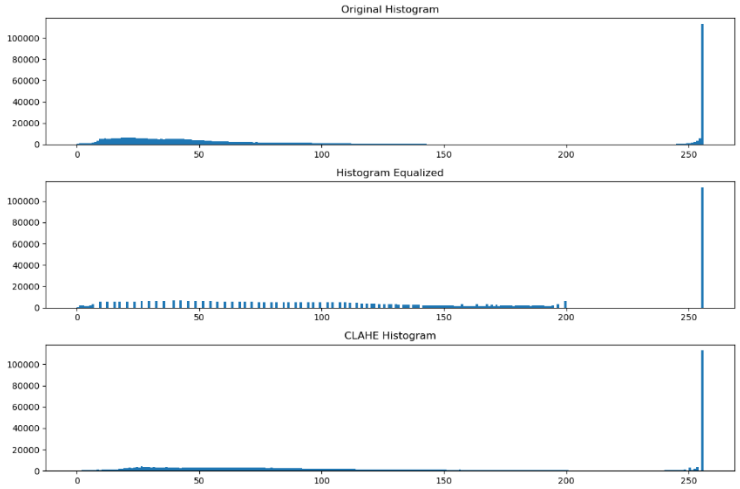
plt.subplot(2, 2, 3)
plt.imshow(hist_equalized, cmap="gray")
```

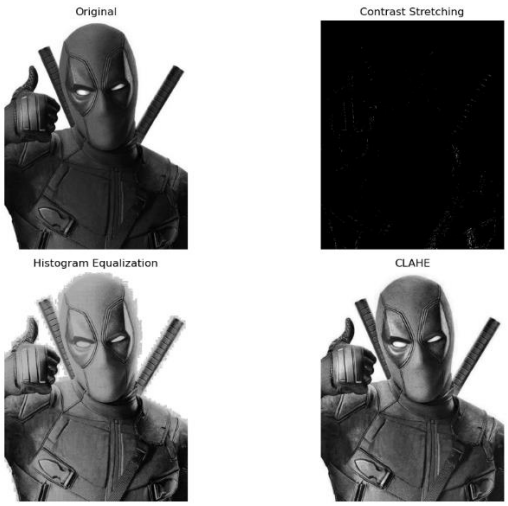
```
plt.title("Histogram Equalization")
plt.axis("off")

plt.subplot(2, 2, 4)
plt.imshow(clahe_img, cmap="gray")
plt.title("CLAHE")
plt.axis("off")

plt.tight_layout()
plt.show()
```

OUTPUT-





QUESTION AND ANSWER

1. What is image contrast, and why is it important in image processing?

Answer: Image contrast is the difference in intensity between the bright and dark regions of an image. High contrast makes objects and details more visible. It is important for image enhancement, segmentation, feature extraction, and object recognition.

2. Explain the concept of an image histogram. What information does it provide?

Answer: An image histogram is a graphical representation of the frequency of different intensity values in an image. It shows whether the image is dark, bright, low-contrast, or high-contrast and helps in selecting suitable enhancement techniques.

3. Differentiate between Histogram Stretching and Histogram Equalization.

Answer: Histogram Stretching: Expands the existing intensity range of an image to improve contrast. Histogram Equalization: Redistributes pixel intensities to produce a more uniform histogram and improve overall contrast.

4. What is Contrast Limited Adaptive Histogram Equalization (CLAHE)? How does it differ from standard Histogram Equalization?

Answer: CLAHE is a local contrast enhancement technique that divides an image into small regions and performs histogram equalization separately. It limits excessive contrast amplification using a clip limit. Standard Histogram Equalization works on the entire image, while CLAHE enhances local details and reduces over-enhancement.

5. Why is histogram equalization commonly applied before feature extraction and image segmentation?

Answer: Histogram equalization improves contrast and visibility of important features. This makes edges, textures, and object boundaries clearer, allowing feature extraction and segmentation algorithms to work more effectively.

6. Mention three real-world applications where histogram equalization is widely used.

Answer: 1Medical imaging – enhancing X-rays, CT scans, and MRI images.2 Satellite/remote sensing – improving geographical and terrain images.3 Surveillance and security – enhancing low-light camera images.

7. What are the limitations of global histogram equalization?

Answer: Global histogram equalization can over-enhance noise, produce unnatural-looking images, and may fail to improve details in specific regions when illumination varies across the image.

8. How does contrast enhancement improve the performance of object detection and recognition systems?

Answer: Contrast enhancement makes objects, edges, and important features more distinct from the background. This provides better input for detection and recognition algorithms, improving their accuracy, especially in poor-light conditions.

9. Compare histogram-based enhancement techniques with brightness adjustment methods.

Answer: Histogram-based techniques: Modify the distribution of pixel intensities to improve contrast and reveal details. Brightness adjustment: Mainly shifts pixel intensity values to make an image lighter or darker. Histogram methods generally improve contrast, while brightness adjustment mainly changes overall illumination.

10. Why is CLAHE preferred for medical imaging and low-light image enhancement?

Answer: CLAHE enhances local details while limiting excessive contrast amplification. It is useful for medical and low-light images because it can reveal subtle features while reducing noise amplification and avoiding over-enhancement.